Transfer Learning Strategies for Jet Tagging at the LHC with Pretrained Foundation Models

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ABSTRACT

Large machine learning models pretrained on simulated particle jets are increasingly used in particle physics, but training these models from scratch is computationally expensive. We measure performance scaling curves for the Sophon foundation model on a 10-class jet classification benchmark, testing three adaptation strategies: frozen backbone, partial fine-tuning, and full fine-tuning. We use nine dataset sizes from 10,000 to 100,000,000 labeled jets. All three strategies resemble power-law improvement curves across the full data range, reaching similar performance but differing in efficiency at small dataset sizes. A frozen Signature-Oriented Pre-training for Heavy-resonance Observation (Sophon) backbone with only a 35,000-parameter multilayer perceptron (MLP) classification head reaches an area under the curve (AUC) of 0.979 at 100M jets, reaching a performance of 99.08% of the published state-of-the-art Particle Transformer (ParT) AUC, while the MLP only uses 1.63% of the trainable parameters of ParT. These results provide guidance for choosing adaptation strategies and show that a frozen pretrained backbone with a small task-specific head is sufficient for most jet tagging tasks. We also provide transfer learning scaling for full finetuning and partial finetuning of the Sophon backbone with the MLP head, that perform marginally better at larger dataset sizes.

INTRODUCTION

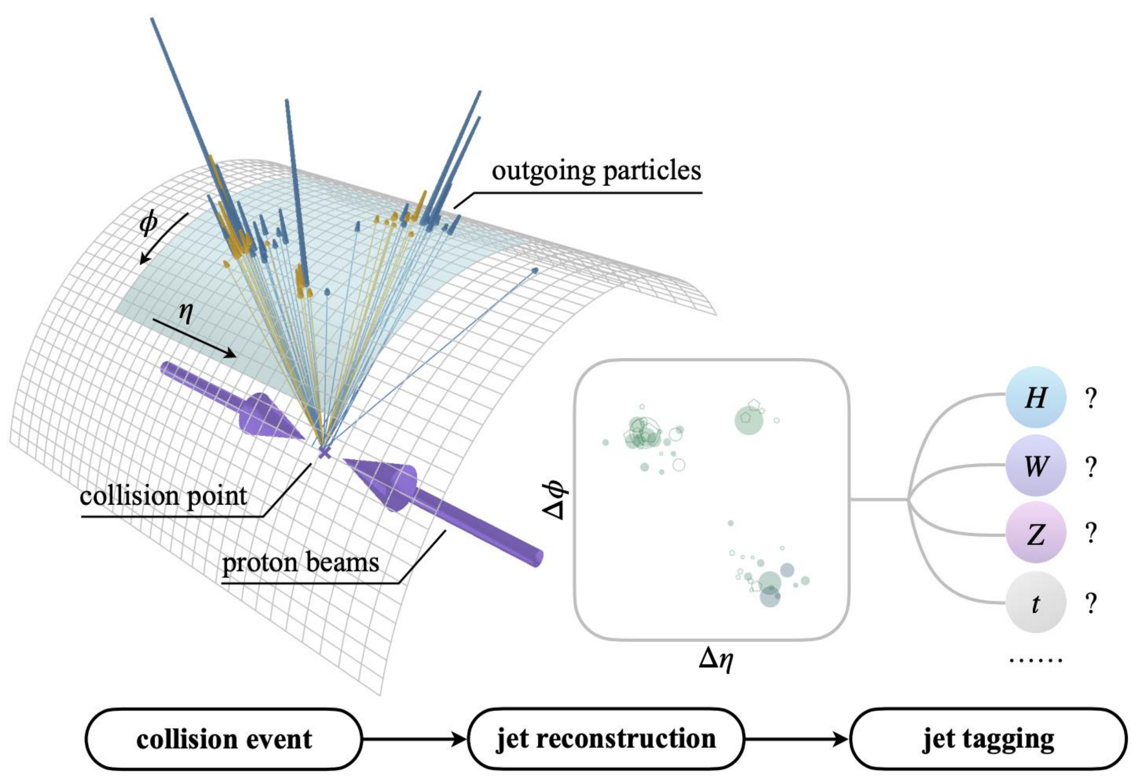

Fig 1: Proton-proton collison at the LHC. From Qu, Li & Qian, Particle Transformer for Jet Tagging, ICML 2022 (arXiv:2202.03772).

- The Large Hadron Collider (LHC) collides proton bunches together ~40 million times per second. Heavy particles (Higgs, W, Z, top quarks) break apart into cone-shaped sprays of lighter particles forming jets
- Jet tagging is identifying which heavy particle created a jet, based on the spray pattern of its lighter decay products
- This is essential for discovering new particles, and a major computing bottleneck for offline analysis
- Transfer learning reuses a pretrained model's knowledge, and allows for adaptation to other datasets

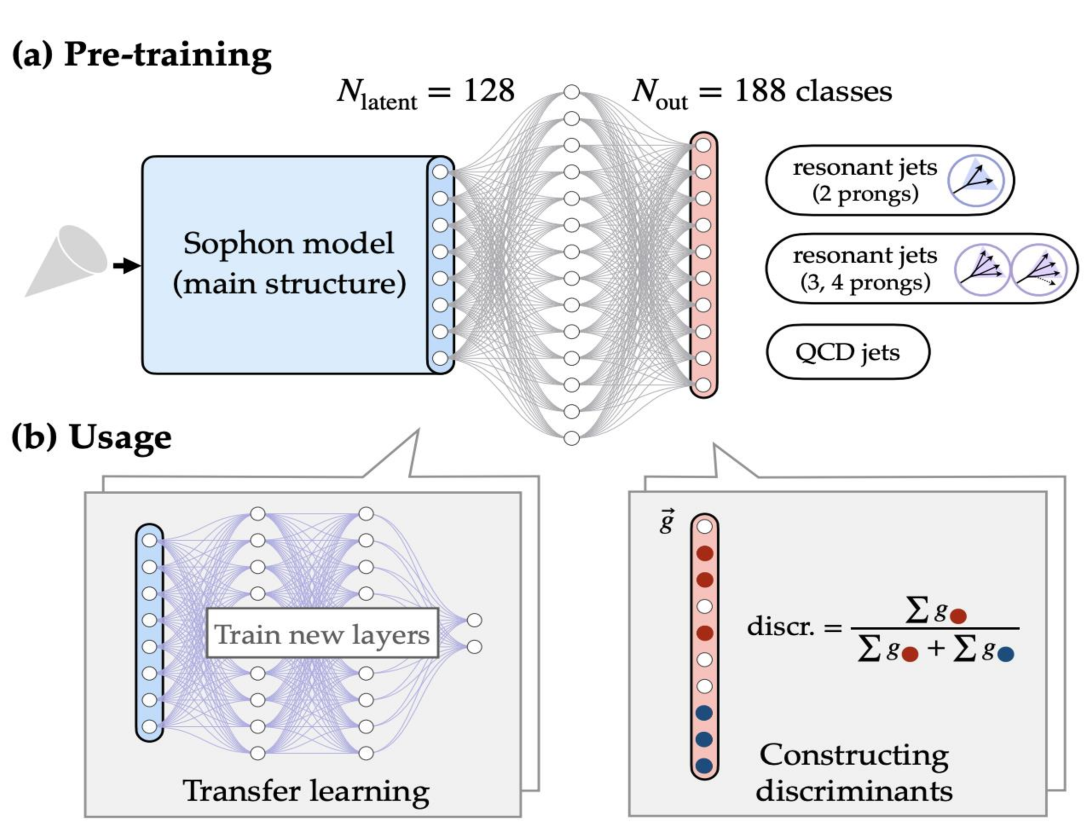

Fig 2: From Li, Du, Qu, Lai et al., Accelerating Resonance Searches via Signature-Oriented Pre-training, arXiv:2405.12972 (2024).

- Sophon is pretrained on 188 jet classes (JetClass-II dataset, ~250M jets) using large-scale supervised learning
- Backbone: Particle Transformer with 8 attention blocks, 2.2M parameters, producing a 128-number embedding per jet
- We adapt a general understanding of jet physics to a simpler 10-class task (JetClass Dataset)

METHODS

Three adaptation strategies applied to the pretrained Sophon backbone (128-dimensional jet embedding → 10-class classification head):

i. Frozen Backbone + MLP: Freezes all Sophon weights, only trains an MLP that maps the 128-number embedding to 10 classes. 35K trainable parameters.
ii. Partial Fine-Tune: Freezes the first 4 layers of Sophon and finetunes the last 4 layers and the MLP. 1.4M trainable parameters.
iii. Full Fine-Tune: Fine tunes Sophon weights and the MLP. 2.2M trainable params.

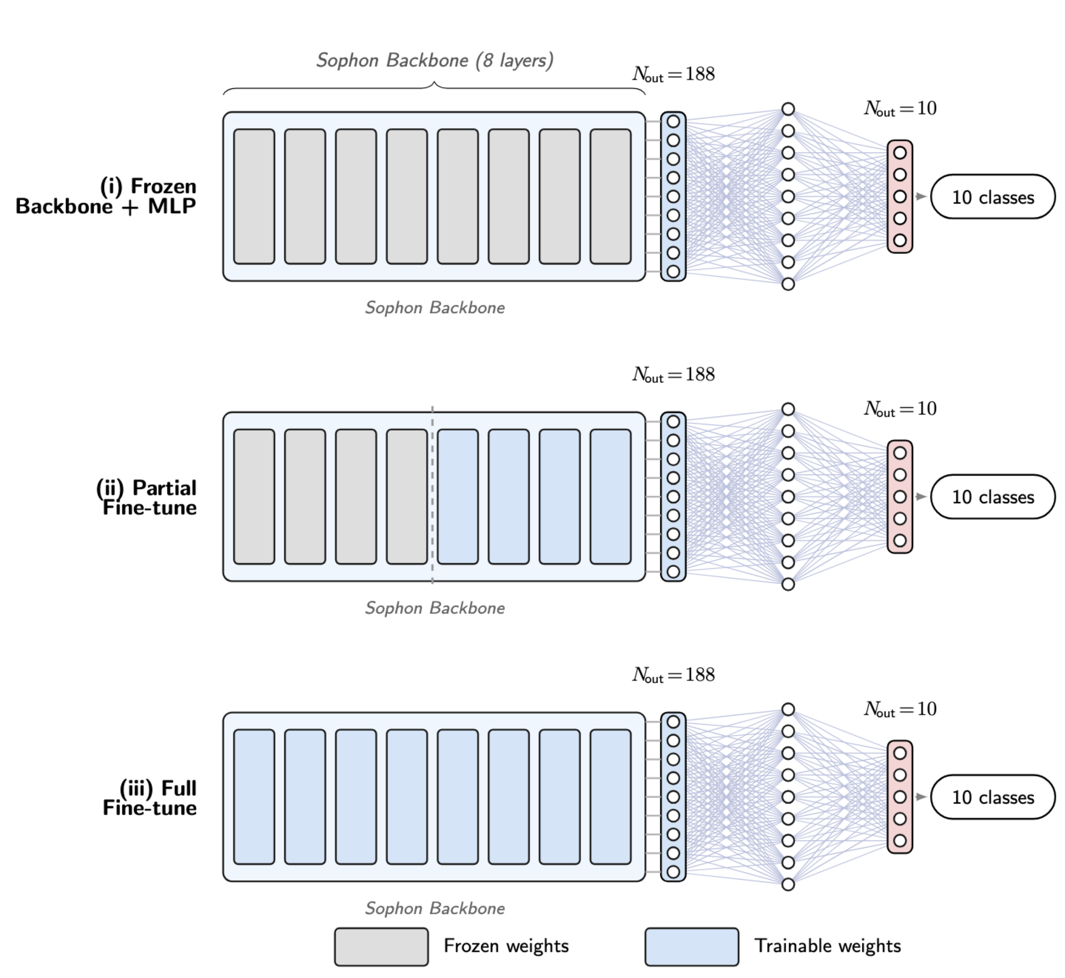

Fig 3: Three Implementations: Frozen Backbone + MLP, Partial Fine Tune, Full Fine-Tune

RESULTS

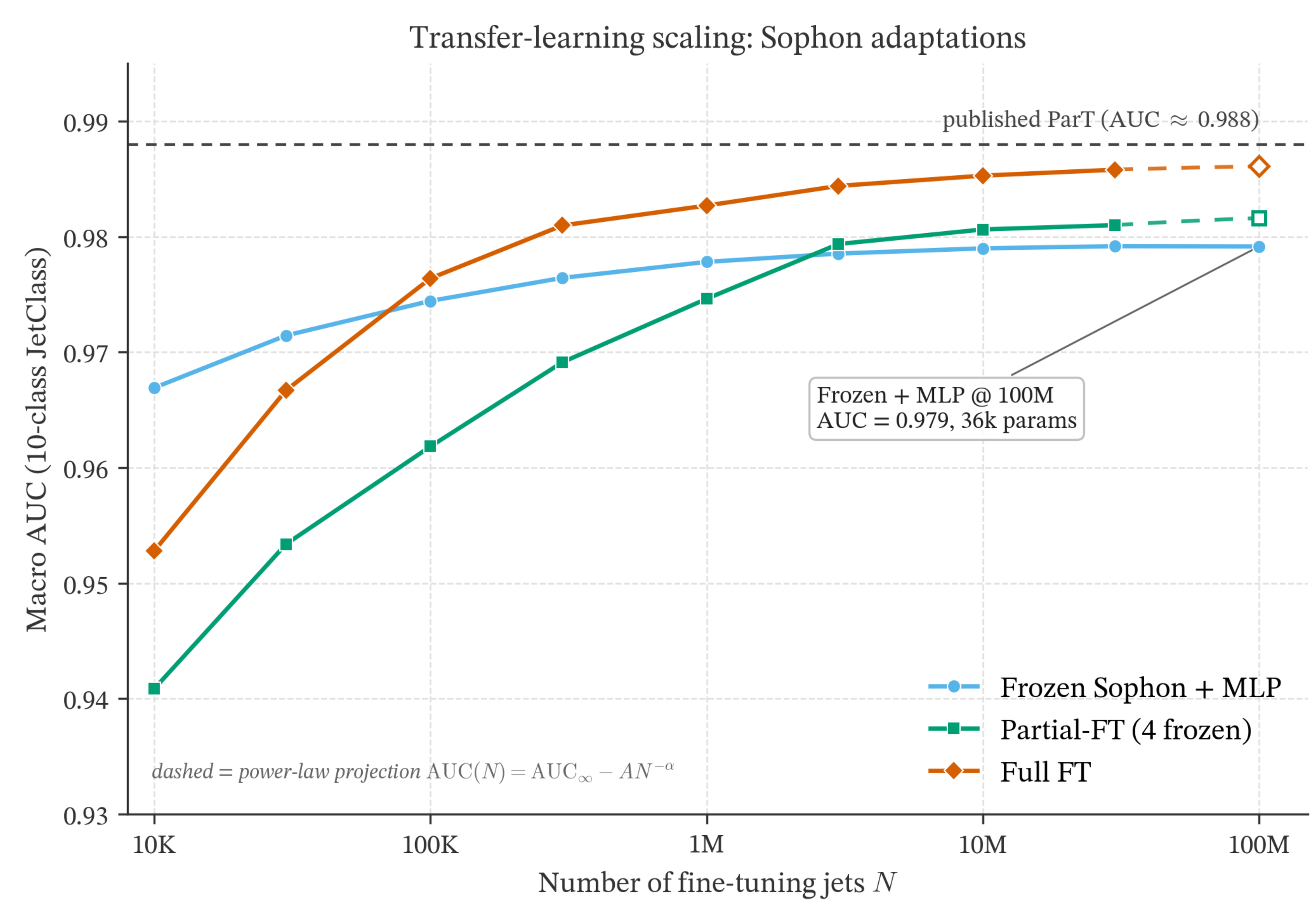

Fig 4: Transfer learning scaling demonstrating Number of fine-tuning jets vs. AUC

- Each curve represents the test AUC, with dataset sizes from 10K to 100M jets
- Solid markers are measured values, and dashed lines are power-law best fits: $AUC(N) = AUC_\infty - A \cdot N^{-\alpha}$
- Frozen Sophon + 35K MLP reaches an AUC of 0.979, reaching 99.08% of the state-of-the-art's AUC of 0.988, using only 1.63% of the trainable parameters.
- The crossover at ~90K jets marks where full fine-tuning begins to outperform frozen feature reuse

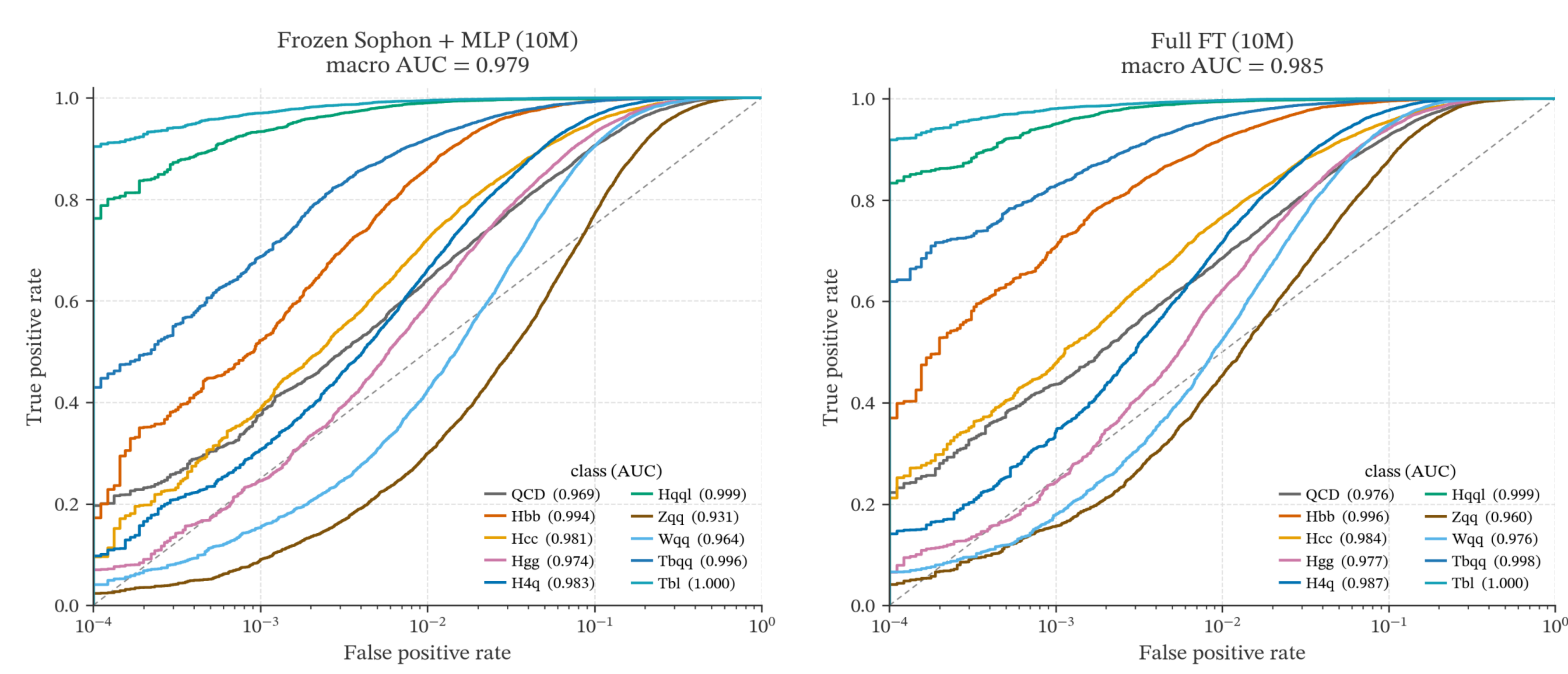

Fig 5: AUC scores for Full FT and Frozen Sophon + MLP

- 10 class-vs-rest ROC curves in each panel (log-scale false-positive axis)
- Macro AUC: Frozen 0.979, full fine-tune 0.985.
- Heavy-flavor jets (H→bb, t→bqq) are easiest to identify, reaching AUC > 0.99; gluon-tagging (separating g from H→gg, H→cc) is hardest. This is consistent with known physics

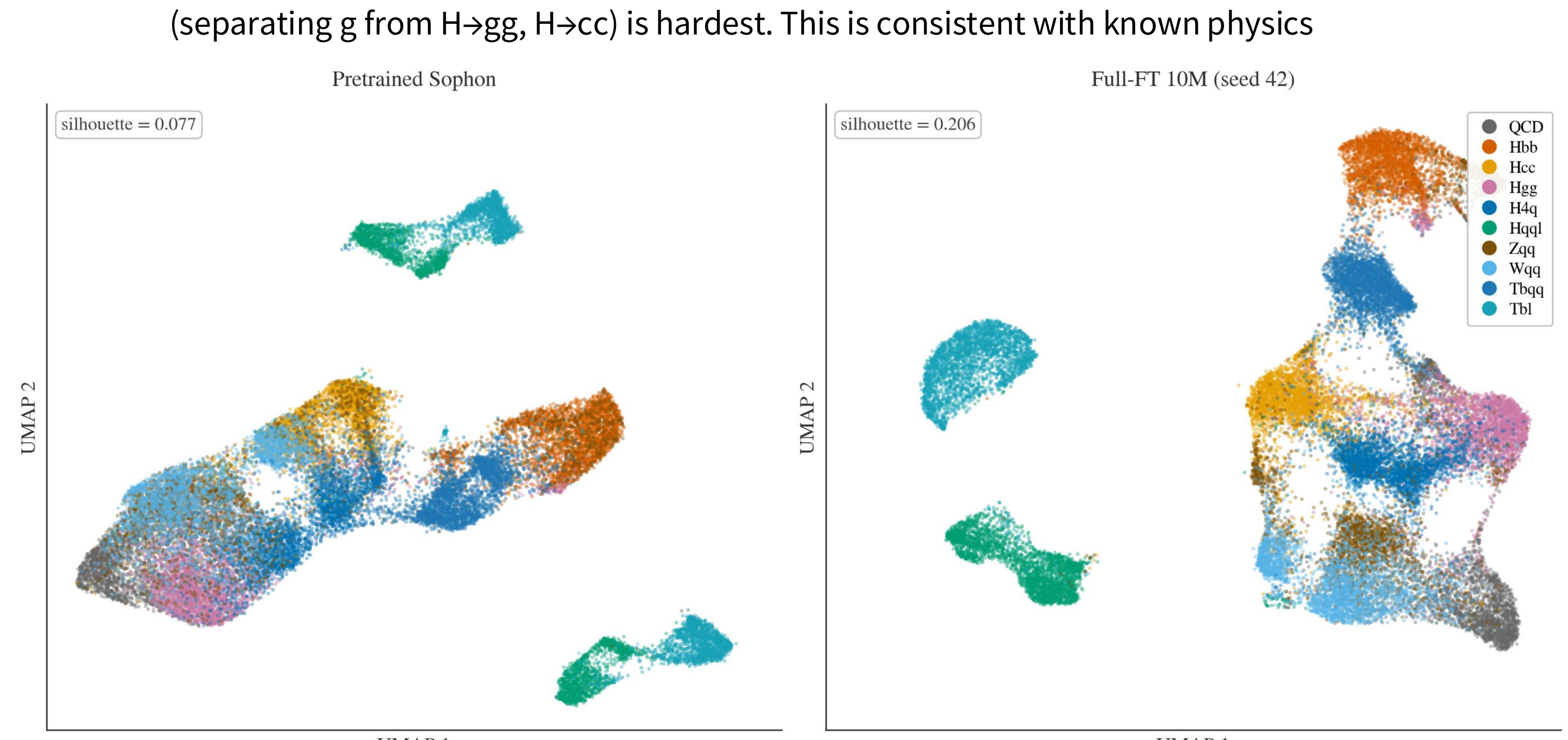

Fig 6: 2D projections of Sophon's 128-dimensional jet embeddings

- 2D projections of Sophon's 128-dimensional jet embeddings for 30K test jets at 10M jets
- Even the pretrained backbone (left) already groups jet types into visible clusters; full fine-tuning (right) makes those clusters much tighter and more separated
- Silhouette score, representing seperation, improves 0.077 → 0.206, improving by a factor of ~2.7

CONCLUSION

- A 35,000-parameter classification head trained on top of a frozen Sophon backbone reaches AUC 0.979, at 99.08% of the Particle Transformer's published 0.988, at 1.63% of the trainable parameters and with zero backbone updates.
- Full fine-tuning extracts a small additional gain, but at a substantially higher computing cost.
- This demonstrates the effectiveness of the MLP with frozen Sophon embeddings with less data. Partial or full fine-tuning is useful when labeled data is abundant and a higher AUC matters for the downstream physics analysis.